

Benchmarking PDE Priors in SIREN-Based PINNs for Zero-Shot Single-Image Super-Resolution

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Abstract—Single-image super-resolution can be written as an ill-posed inverse problem in which many high-resolution images are consistent with the same low-resolution observation. While supervised super-resolution networks achieve strong visual quality, they require paired datasets and encode the image prior implicitly. This work instead focuses on an interpretable zero-shot setting and documents a benchmark built on physics-informed neural networks (PINNs). The unknown high-resolution image is represented by a continuous SIREN and constrained by a known degradation operator composed of Gaussian blur and subsampling. Within this unified framework we compare classical PDE-inspired priors, namely Perona-Malik diffusion and vectorial anisotropic diffusion, each coupled with either Total Variation or shock regularization. The benchmark keeps the backbone, optimization schedule, warm start, and synthetic degradation pipeline fixed so that only the regularization strategy changes across trials. The final evaluation is designed on the Set5 Baby and Butterfly images at $\times 2$ and $\times 4$, with an additional small out-of-distribution check on up to two Darmstadt images. This draft reports the full methodology, implementation details, and result templates ready to be completed once the final runs are available.

Index Terms—physics-informed neural networks, single-image super-resolution, SIREN, anisotropic diffusion, total variation, benchmark

I. INTRODUCTION

Single-image super-resolution (SISR) aims at recovering a high-resolution (HR) image from a single low-resolution (LR) observation. The problem is intrinsically ill-posed because the forward operator discards high-frequency information, so infinitely many HR images can explain the same LR input. Modern supervised approaches such as SRCNN, EDSR, SwinIR, and diffusion-based models obtain strong performance by learning an implicit prior from large paired datasets [8]–[11]. However, they require offline training data and offer limited control over the physical consistency of the reconstruction.

Classical PDE-based regularizers provide a complementary viewpoint. Perona-Malik diffusion, anisotropic diffusion, Total Variation (TV), and shock filters were originally introduced for denoising and edge-preserving restoration [4]–[7]. They are interpretable, modular, and well aligned with inverse problems, but they are usually implemented on fixed grids and are less commonly benchmarked inside continuous neural representations.

This paper documents a controlled benchmark where the HR image is represented by a SIREN [2] and regularized by PDE-inspired losses inside a PINN framework [1]. The emphasis is on benchmarking, not on proposing a new architecture. The same backbone, degradation model, optimizer, and warm start are kept fixed, while the regularizer is changed systematically. This setup isolates the effect of scalar versus vectorial diffusion and of TV versus shock sharpening in a zero-shot regime.

The main contributions of the proceeding are the following.

- A unified PINN formulation for zero-shot SISR with a continuous SIREN parameterization and a known blur-plus-subsampling forward operator.
- A benchmark matrix that compares diffusion priors in two branches: diffusion plus TV and diffusion plus shock, under identical optimization settings.
- A reproducible protocol on Set5 Baby and Butterfly at $\times 2$ and $\times 4$, plus a small external check on two Darmstadt images used as additional HR sources before synthetic degradation.
- A complete $\LaTeX$ draft with sections, tables, and figure placeholders that can be filled directly once the final runs and training plots are produced.

The remainder of the paper is organized as follows. Section II reviews the closest literature. Section III formulates the inverse problem. Section IV describes the PINN benchmark.

Section V details the implementation choices. Section VI defines the final protocol and the result templates. Section VII summarizes the planned analysis, and Section VIII concludes the paper.

II. RELATED WORK

A. Data-Driven Super-Resolution

Deep learning has dominated SISR for the last decade. Early CNN models such as SRCNN introduced direct end-to-end mapping from LR to HR [8]; residual and transformer architectures such as EDSR and SwinIR pushed the quantitative frontier further [9], [10]; diffusion-based pipelines improved perceptual realism [11]. These methods are highly effective when a representative training distribution is available, but their prior is learned offline and can be difficult to adapt when the degradation operator is known and the data regime is extremely small.

B. PDE-Based Image Restoration

PDE-based models remain attractive because they encode restoration assumptions explicitly. Perona-Malik diffusion suppresses smoothing across strong gradients [4]; anisotropic diffusion extends this idea with directional control [5]; TV promotes sparse gradients and piecewise-smooth reconstructions [6]; shock filtering sharpens edges through a nonlinear transport mechanism [7]. These priors were developed mainly for denoising and deblurring, yet they are natural candidates for the free subspace of an ill-posed SR inverse problem.

C. PINNs and Implicit Representations

PINNs enforce physical constraints through differentiable residuals computed by automatic differentiation [1]. For image restoration, they provide a convenient bridge between continuous representations and PDE priors. SIREN networks are especially suited to this setting because sinusoidal activations represent fine detail and yield stable first- and second-order derivatives [2]. The repository used in this work also contains a Fourier-feature MLP alternative [3], but the benchmark described here fixes the backbone to SIREN in order to isolate the role of the regularization terms.

III. PROBLEM FORMULATION

A. Forward Model

Let $u \in [0, 1]^{H \times W \times 3}$ be the unknown HR image and $y \in [0, 1]^{h \times w \times 3}$ the observed LR image. The degradation model implemented in the codebase is

$$y = Pu + \eta, \quad P = SB, \quad (1)$$

where B is a separable Gaussian blur with standard deviation σ , S is stride-based subsampling with factor s , and η is optional additive noise. In the current benchmark, the core SR setting uses $\sigma = 1.0$, $s \in \{2, 4\}$, and zero added noise so that the comparison focuses on the regularizer rather than on noise robustness.

B. Inverse Problem

The reconstruction is formulated as

$$u^* = \arg \min_u \|Pu - y\|_2^2 + \lambda R(u), \quad (2)$$

where $R(u)$ is a prior that biases the solution toward plausible HR images. In the proposed PINN setting, u is not optimized directly on a discrete grid. Instead, it is represented by a neural function $u_\theta : [0, 1]^2 \rightarrow \mathbb{R}^3$ and the prior is enforced through differentiable residual terms sampled at collocation points. This allows the inverse problem to remain continuous while still using classical PDE operators.

IV. BENCHMARK METHOD

A. Continuous Image Representation

The benchmark fixes the HR image model to a SIREN [2]. Given normalized coordinates $x \in [0, 1]^2$, the network outputs RGB values $u_\theta(x) \in [0, 1]^3$. The implementation uses five layers, hidden size 256, and first-layer frequency parameter $\omega_0 = 30$. This representation is shared by all trials.

B. Data Fidelity

The main fidelity term in the repository is

$$\mathcal{L}_{\text{data}}(\theta) = \|Pu_\theta - y\|_2^2, \quad (3)$$

implemented by rendering the full HR grid, applying the same blur-plus-subsampling operator used to synthesize the LR observation, and computing the mean squared error. A second pointwise term exists in the codebase, but it is not part of the core benchmark so that every configuration is anchored by the same physically meaningful $\mathcal{L}_{\text{data}}$.

C. PDE and Sharpness Priors

The benchmark compares two diffusion priors and two sharpening regularizers.

Perona-Malik diffusion. For each channel,

$$N_{\text{PM}}[u] = \text{div}(g(\|\nabla u\|^2)\nabla u), \quad g(s) = \frac{1}{1 + s/\kappa^2}. \quad (4)$$

This is the scalar edge-preserving baseline.

Vectorial anisotropic diffusion. RGB channels are coupled through the structure tensor

$$G(u) = \sum_{c=1}^3 \nabla u_c \nabla u_c^\top. \quad (5)$$

Let $G = Q \text{diag}(\lambda_1, \lambda_2) Q^\top$ and define $D(u) = Q \text{diag}(g_1, g_2) Q^\top$ with $g_i = 1/(1 + \lambda_i)$. The residual is

$$N_{\text{aniso}}[u]_c = \text{div}(D(u)\nabla u_c). \quad (6)$$

The current implementation evaluates $G(u)$ pointwise at the sampled collocation coordinates and regularizes it before eigendecomposition.

Total Variation. The vectorial TV loss is

$$\mathcal{L}_{\text{TV}}(\theta) = \mathbb{E}_x \left[\sqrt{\sum_{c=1}^3 \|\nabla u_c(x)\|^2 + \varepsilon} \right]. \quad (7)$$

It promotes sparse gradients and therefore piecewise-smooth images with sharper edges.

Shock filter. Let $\eta = \nabla u / (\|\nabla u\| + \varepsilon)$ denote the local gradient direction. The implemented differentiable shock residual is

$$N_{\text{shock}}[u] = \tanh\left(\frac{u_{\eta\eta}}{\varepsilon}\right) \|\nabla u\|, \quad (8)$$

where $u_{\eta\eta}$ is the second directional derivative along η . This term emphasizes edge sharpening but is potentially more aggressive than TV.

D. Boundary Conditions and Composite Loss

Zero normal derivative boundary conditions are enforced on the four borders of the normalized square domain. The total loss is

$$\mathcal{L} = \lambda_d \mathcal{L}_{\text{data}} + \alpha(t) \sum_{j \in \mathcal{P}} \lambda_j \mathcal{L}_j + \lambda_{bc} \mathcal{L}_{bc}, \quad (9)$$

where $\mathcal{P}$ contains the active prior terms and $\alpha(t)$ is a linear curriculum that ramps PDE-type contributions from 0 to 1 after an initial warm-up. The benchmark will compare priors under this same loss template.

V. IMPLEMENTATION DETAILS

All experiments use the same settings taken from the current codebase. HR images are resized to 256×256 before degradation. The training loop runs for 2000 epochs with Adam, learning rate 10^{-4} , 4096 collocation points for prior terms, 2048 sampled LR data points, and gradient clipping at norm 1.0. Before the main optimization, the network is warm-started for 2000 steps to reproduce the bicubic upsampling of the LR image using Adam with learning rate 10^{-3} and cosine annealing. This shared initialization is important because it reduces poor local minima and makes cross-method comparisons fairer.

Two numerical details are essential. First, derivatives are computed on coordinates normalized to $[0, 1]^2$, so second-order residuals scale with the square of the image size. The implementation therefore divides PDE residuals by $\max(H, W)^2$ to keep the loss magnitudes comparable across resolutions. Second, the anisotropic tensor adds εI before eigendecomposition to avoid numerical instability in flat regions. The current reference weights are $\lambda_{\text{PM}} = \lambda_{\text{aniso}} = 2 \times 10^{-2}$, $\lambda_{\text{TV}} = 2 \times 10^{-3}$, $\lambda_{\text{shock}} = 5 \times 10^{-3}$, and $\lambda_{bc} = 10^{-3}$, with PDE warm-up starting after 50 epochs and ramping over the next 200 epochs.

The repository also contains fourth-order terms and a learned CNN prior. These components are deliberately excluded from the main benchmark described in this paper, whose final comparison is restricted to the SIREN backbone and to second-order diffusion priors combined with TV or shock. Preliminary fourth-order sweeps are treated as exploratory only and serve to motivate later follow-up experiments rather than the core table of the proceeding.

TABLE I
PLANNED BENCHMARK CONFIGURATIONS.

ID	Active terms	Goal
B0	data	data-only PINN baseline
T0	data + TV	sharpness-only baseline
T1	data + PM + TV	scalar diffusion + TV
T2	data + Aniso + TV	vectorial diffusion + TV
S0	data + Shock	sharpness-only baseline
S1	data + PM + Shock	scalar diffusion + shock
S2	data + Aniso + Shock	vectorial diffusion + shock

VI. EXPERIMENTAL PROTOCOL

A. Final Benchmark Matrix

Table I defines the configurations planned for the final benchmark. The matrix follows the latest project direction: compare the same data term against two regularization branches, diffusion plus TV and diffusion plus shock.

If the additional ZG-style candidate remains stable after the closing exploratory runs, one extra row can be appended to each branch without changing the rest of the protocol. In the codebase, these rows map to `data_lr`, `reg_tv`, `pde_perona_malik`, `pde_anisotropic`, and `pde_shock`.

B. Images, Scales, and Metrics

The main evaluation is planned on the Set5 images `baby.png` and `butterfly.png`. For each image, the same synthetic degradation pipeline is applied and the methods are tested at scales $s \in \{2, 4\}$. Results will be reported in terms of PSNR and SSIM, matching the metrics already implemented in the repository. The comparison against bicubic interpolation is retained as the main classical baseline, while the data-only PINN baseline B0 shows how much is gained by the regularizer beyond the inverse constraint itself.

As a small out-of-distribution check, up to two additional images originating from the Darmstadt benchmark will be used as external HR sources before synthetic degradation. These extra runs are not intended to define a new leaderboard; they only test whether the ranking of the priors is stable outside the two Set5 reference images.

C. Planned Outputs

Besides the quantitative tables, the final paper will include:

- qualitative crops on Baby and Butterfly at $\times 2$ and $\times 4$;
- training curves for data, PDE, and boundary terms, sampled every fixed number of epochs;
- a short stability analysis comparing the convergence behavior of TV-based and shock-based branches.

The current trainer already produces snapshots every 25 epochs, so the loss-curve figure can follow the same cadence when the logging script is completed.

TABLE II
TEMPLATE FOR THE FINAL AVERAGE RESULTS ON SET5 BABY AND BUTTERFLY.

Method	$\times 2$		$\times 4$	
	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
Bicubic	—	—	—	—
B0	—	—	—	—
T0	—	—	—	—
T1	—	—	—	—
T2	—	—	—	—
S0	—	—	—	—
S1	—	—	—	—
S2	—	—	—	—

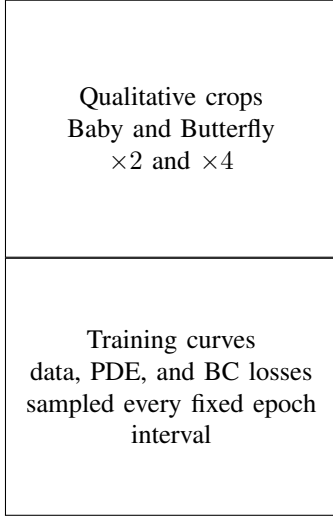


Fig. 1. Placeholders for the final qualitative and convergence figures. The draft compiles without external assets and keeps the layout ready for the final plots.

VII. PLANNED ANALYSIS

The benchmark is designed to answer a small set of focused questions. First, does vectorial anisotropic diffusion improve over scalar Perona-Malik diffusion when the RGB channels are coupled through a shared structure tensor? This is particularly relevant near chromatic edges, where channel-wise diffusion can produce color fringing. Second, what is the trade-off between TV and shock regularization once the same data term and the same diffusion prior are fixed? TV is expected to be smoother and more stable, whereas shock-based variants may recover crisper transitions at the price of higher sensitivity.

Third, how much do the gains depend on the difficulty of the inverse problem? The $\times 2$ setting is relatively mild, while the $\times 4$ setting leaves a larger null space to the prior. This makes the pair $\{\times 2, \times 4\}$ informative even with only two core images. Finally, the two Darmstadt-based runs will act as a small sanity check on whether the qualitative ranking of the priors is tied too strongly to Set5.

VIII. CONCLUSION

This draft consolidates the methodological part of a benchmark on PDE-inspired priors for zero-shot single-image super-resolution with SIREN-based PINNs. The paper now contains the full problem statement, the exact benchmark matrix, the implementation details needed for reproducibility, and the tables and figures required for the final write-up. Once the last experimental runs are completed, the remaining work is limited to filling the quantitative tables, inserting the qualitative crops, and adding the convergence plots.

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